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ML Course
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Sec 22
Inferential Statistics

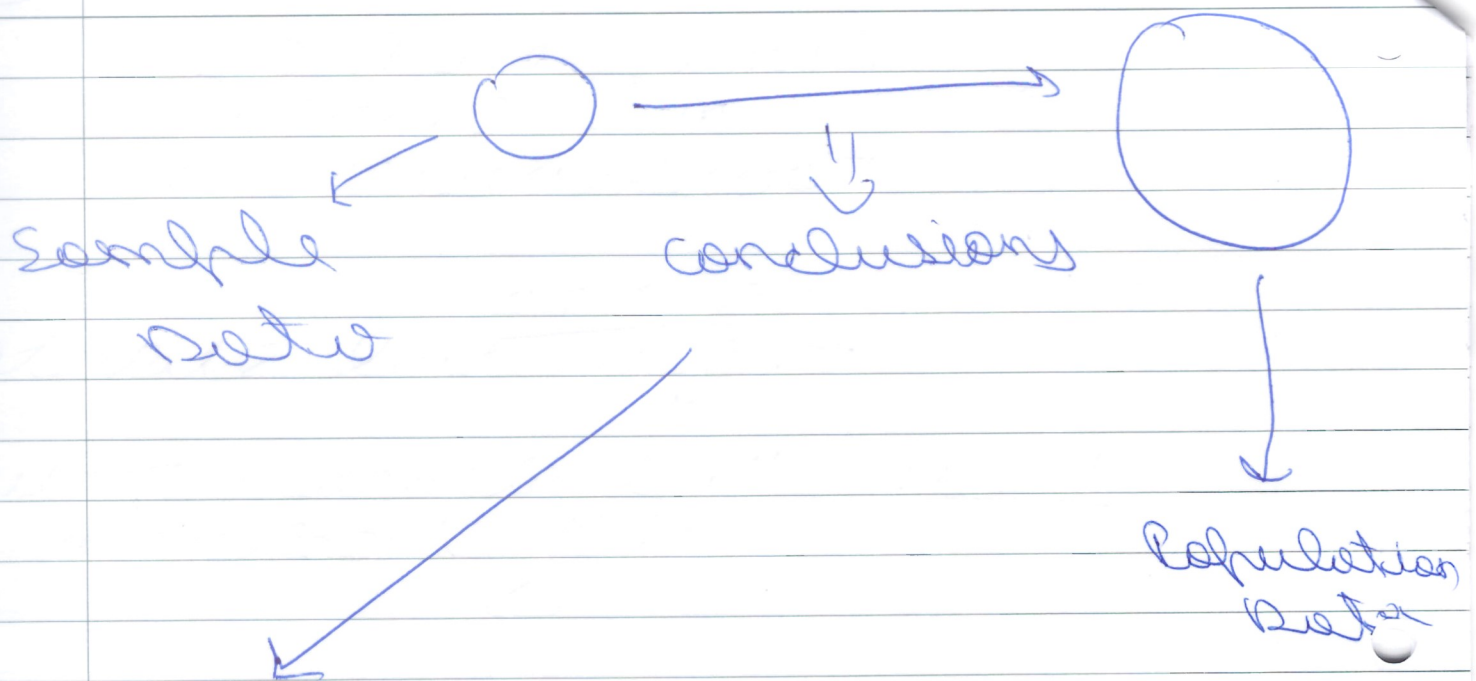
Sec-1:- Hypothesis testing
and mechanism
↓

Sec-2:- What is
P Value?

A Hypothesis and
Hypothesis testing
mechanism.

Inferential Statistics:-

→ Drawing conclusion
or Inferences from
unknown sample data



For the conclusions or inferences, we use the Hypothesis testing.

=> Hypothesis testing mechanism

① Null Hypothesis (H_0)

=> the assumptions you are beginning with.

② write your alternate hypothesis.

③ A Alternate hypothesis is opposite of null hypothesis.

(3) Then we ~~are~~ perform experiments which are basically statistical analysis.

→ Here we collect data.

(4) Accept the Null Hypothesis or reject the Null Hypothesis.

Q.1. Colleges in district A reported the average pass % of students is 85%.

A new college opened in district A a sample of 40% of that school had pass % of 90%.

with std. deviation of 47.

P. 7.5

Does school have a different pass %?

Soln

(1) First define null hypothesis

$(H_0) \Rightarrow \mu = 85\%$ (Pass percentage)

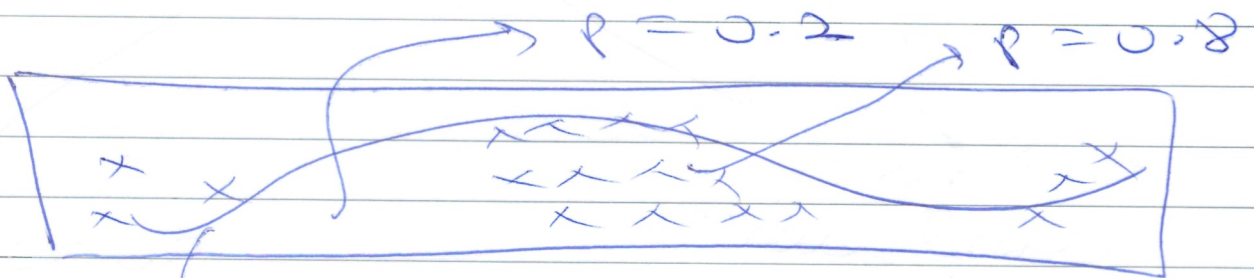
Alternate hypothesis $(H_1) \Rightarrow \mu \neq 85\%$

$\rightarrow$ Hence not equal to 85%

$\Rightarrow$ P value

The p value is a number calculated from a statistical test that describes that how likely you are to see some ground in particular set of observed events, if the null hypothesis were true

- (★) The tower distribution is like a Gaussian Normal distribution.



Sphere Ball

Here is the Gaussian distribution shown

Out of 100 towers in sphere ball, we touch around 20 times in this region.

(★) Hypothesis Testing :-

Ex :- Coin is fair or not.

→ we have to check this

Suppose $P(H) = 0.5$ $P(T) = 0.5$

⑤

Suppose if we toss
100 times

50 times
tail will
come

50 times
heads will
come

∴ coin is fair.

Ex - Coin is fair or not
{ 10 times }

$$P(H) = 0.5 \quad P(T) = 0.5$$

$$P(H) > 0.5 \quad P(T) = 0.4$$

$$P(H) = 0.7 \quad P(T) = 0.3$$

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(1) Null Hypotheses
 H_0 = coin is fair

(2) Alternative Hypotheses
 H_1 = coin is unfair

Suppose in an experiment we have the test is repeated 100 times.



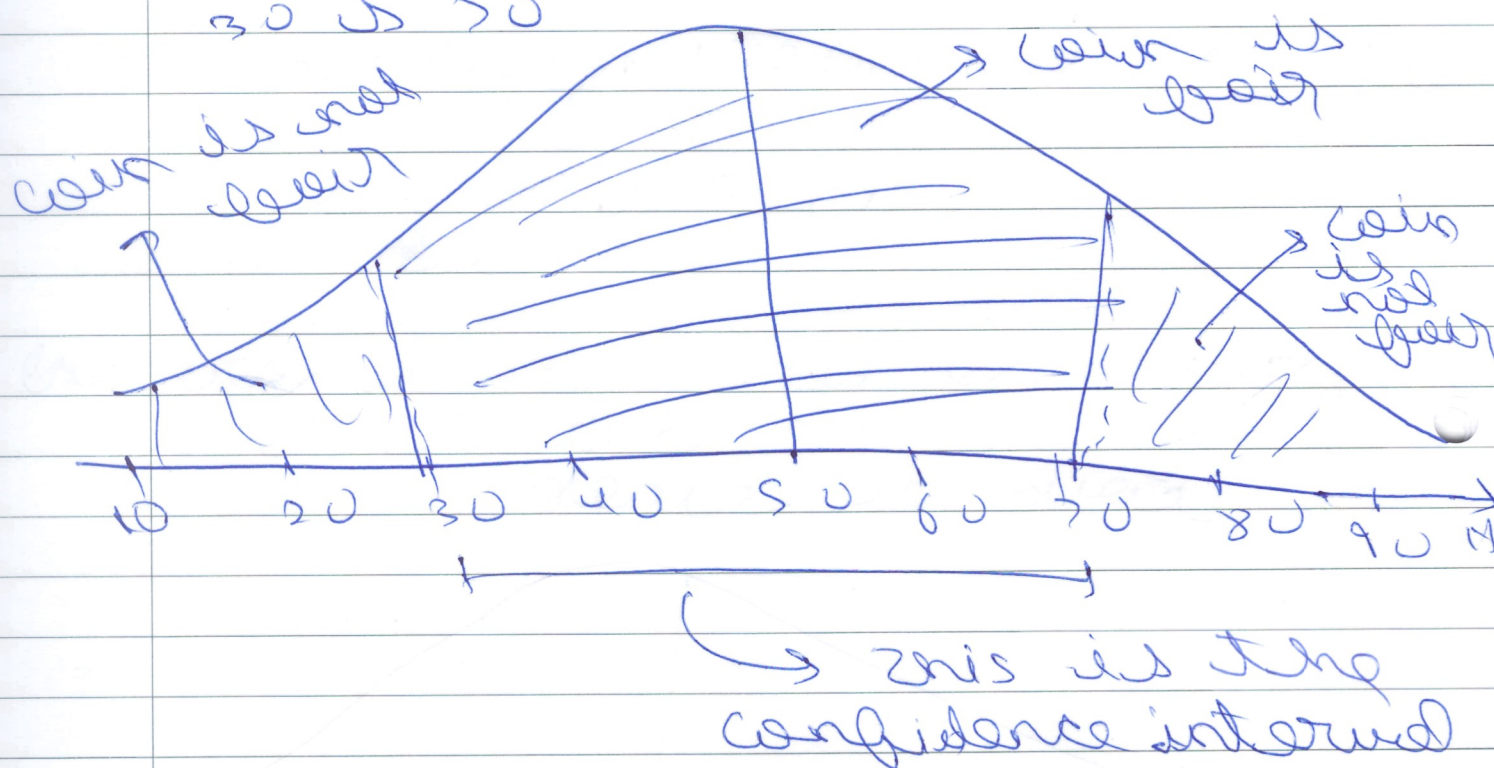
Suppose, the no. of heads is 50 $\Rightarrow$ A fair coin

Suppose we get no. of heads 100

$\hookrightarrow$ then also the coin can be fair

But if heads is 70 times $\hookrightarrow$ then there may be some problem.

④ so now, we define our confidence interval as $30 < \mu < 70$



coin is fair in confidence interval

⑤ Now we do our

there is a very important term known as significance value.

(A)

Significance Value:-

$$\alpha = 0.05$$

Confidence Interval =

$$CI = 1 - 0.05 \\ = 0.95$$

(B)

Suppose the null hypothesis is rejected at 30 degree.

Suppose $P < \text{Significance Value}$

↙
We reject the null hypothesis
also we will reject the null hypothesis